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Unintentional injuries, Mortality trends, Joinpoint analysis, Public health, Ecuador, External causes.

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Joinpoint Regression Analysis of Unintentional Injury Mortality Trends in Ecuador, 2018–2024

Abstract

Introduction: Unintentional injuries pose a major challenge to global health. In Ecuador, the lack of regular epidemiological updates hinders the development of preventive policies and the urgent optimization of national health system resources. To examine the temporal trends and geographical distribution of unintentional injury mortality in Ecuador between 2018 and 2024.

Methods: A retrospective ecological time-series study was conducted using official death records from the National Institute of Statistics and Censuses. Age-standardized mortality rates (per 100,000 inhabitants) were calculated using the direct method, with the world population as the reference population. Trends were analyzed using Joinpoint regression models with Monte Carlo permutation tests to identify significant inflection points and calculate the APC.

Results: Mortality from unintentional injuries showed a non-significant upward trend (APC 0.78; 95% CI: -9.7 to 12.8). A marked male over-mortality was identified, with a ratio of 3.8:1. Transport accidents were the primary cause of death (58.5%), despite a significant downward trend in both sexes. Geographically, mortality shifted from the Highlands toward the Amazon and Insular regions, with a notable increase in 2024 in Sucumbíos, Zamora Chinchipe, and Galápagos.

Discussion: These pronounced geographic shifts toward the Amazon region, combined with significant gender disparities characterized by a marked excess male mortality, strongly highlight the urgent need for targeted, multisectoral safety interventions and the continuous strengthening of local public health infrastructure to prevent future injuries.

Conclusion: Ecuador maintains significant mortality rates from unintentional external causes, with risk patterns clearly differentiated by gender and province. These results emphasize the need for targeted prevention strategies and improved health information systems.

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1. INTRODUCTION

Unintentional injuries are one of the leading causes of death worldwide; these injuries cause approximately 3.9 to 4 million deaths annually, representing around 6 to 7% of total global mortality; more than 90% of this burden falls on low- and middle-income countries [1,2] . Despite this high mortality rate, the incidence of injuries increased between 1990 and 2021, accompanied by a reduction in mortality [3].

Unintentional injuries are those that occur by accident, without any deliberate intent to cause harm. According to the International Classification of Diseases (ICD), external causes include, among others, traffic accidents, falls, poisoning, drowning, and episodes of suffocation [4,5].

In Latin America and the Caribbean, mortality from unintentional injuries occurs regularly, mainly due to falls and traffic accidents. These are strongly influenced by poverty and social fragility. Young men and marginalized populations bear an unfair burden, which entails high economic costs [6,7].

In South American countries such as Colombia and Peru, unintentional injuries represent a significant public health burden. In Colombia, between 1990 and 2021, age-standardized mortality from traffic injuries fell by nearly 40%; however, road accidents remain among the leading causes of death, especially among men [8]. In Peru, the share of deaths from injuries decreased from 14.4% (13.4–15.3) to 8.8% (8.2–9.4) between 1990 and 2019. By the late 1990s, both intentional and unintentional injuries accounted for approximately 10.8% of all deaths, and as in other low- and middle-income countries, road traffic accidents and other unintentional causes account for a large proportion of the burden in children and adults [9].

In Ecuador, mortality from unintentional injuries is dominated by traffic accidents, with rates of around 18 to 20 deaths per 100,000 people. Between 2011 and 2022, there were 38,355 deaths from traffic accidents, with an average mortality rate of 19.4 per 100,000; the overall trend was slightly downward without statistical significance (annual percentage change of −0.4%) [10].

An analysis by age group examined the trend in mortality from traffic injuries, comparing older adults with younger adults; the results showed that older adults had a higher mortality rate (36.2 per 100,000 inhabitants) than younger adults (16.7 per 100,000 inhabitants)[11].

Currently, the available evidence focuses on specific causes and does not provide an overview of trends in mortality from unintentional injuries. This gap in scientific research limits the ability to implement targeted actions to reduce mortality in line with current population needs.

In this scenario, it is important to analyze trends in unintentional injury mortality in Ecuador in recent years to establish appropriate preventive interventions grounded in scientific evidence, such as road safety campaigns or stricter traffic laws [12]. In a context where violence is continuously on the rise and homicides are projected to become the second leading cause of death by 2024, it is essential to identify and prioritize these issues so that public policies respond in a balanced manner to both problems. Such an approach would help reduce their cumulative impact and strengthen a more equitable and resilient public health system.

The objective of this study is to examine mortality from unintentional injuries in Ecuador between 2018 and 2024, observing variations over time, geographic distribution, and disparities according to age groups and sex.

2.METHODS

A quantitative, descriptive, retrospective study was conducted on mortality from unintentional injuries in Ecuador. The analysis was based on official death records for seven years, from January 1, 2018, to December 31, 2024.

The data come from the Statistical Registry of Deaths of the National Institute of Statistics and Censuses (INEC). The ICD-10 codes related to unintentional injuries were selected [13]. They were classified according to cause: transport accidents (V01-V79), other transport accidents (V80-V89), falls (W00-W19), drowning and suffocation (W65-W84), accidents involving fire and hot substances (X00-X19), poisoning (X40-X49), and other types of accidents (W20-W64, W85-W99, X20-X39, X50-X59).

Standardized rates were adjusted for age using the direct method, with the global population as the reference. Specific rates were also estimated for age groups (0-14 years, 15–24 years, 25–44 years, 45–64 years, and ≥65 years) and gender. Results are presented as rates per 100,000 population.

JoinPoint models were used to study trends. The JoinPoint methodology identifies the year of significant trend change using Monte Carlo permutation tests; it also quantifies each trend using the annual percentage change (APC) and its confidence intervals. An inflection point and statistical significance were set at $p<0.05$ [14].

Data analysis was performed using STATA 15 and JoinPoint 5.4.0.

The study used secondary data from official death records and demographic projections, without including personal information or directly intervening with individuals. Therefore, there was no risk to participants, and informed consent was not required, thereby ensuring confidentiality and compliance with ethical principles in public health research.

3.RESULTS

In Ecuador, 41,591 deaths from unintentional injuries were recorded between 2018 and 2024. Figure 1 shows the age-adjusted mortality rates (per 100,000 inhabitants) due to unintentional injuries in Ecuador during 2018–2024, broken down by total population, men, and women.

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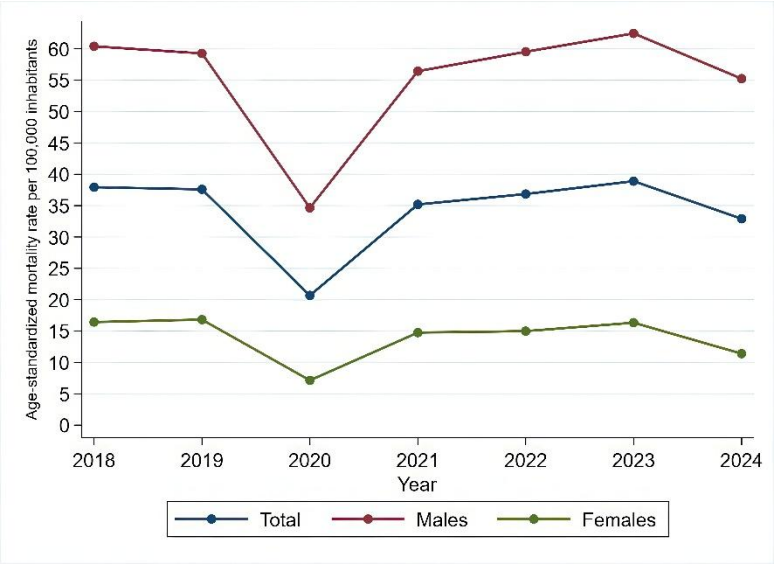


Figure 1. Standardized mortality rates for unintentional injuries in Ecuador, 2018–2024, by sex.

Trend analysis shows a slight increase in unintentional injuries during the study period, with an annual percentage change of 0.78 (95% CI: -9.7; 12.8), which is not statistically significant. In 2018 and 2019, mortality rates remained relatively stable, at 37.94 and 37.60 per 100,000 population, respectively, until 2020, when they fell sharply to 20.67 per 100,000 due to mobility restrictions caused by COVID-19. The increase in mortality rates was observed from 2021 onwards, peaking in 2023 (38.90 per 100,000) and subsequently declining in 2024 (32.91 per 100,000).

During the period analyzed, men accounted for 79.3% of deaths from unintentional injuries, with a male-to-female ratio of 3.8. Mortality rates among men showed a slight but not significant increase, with an annual percentage change of 1.3 (95% CI: -8.4; 12.4). In 2018, the rate reached 60.41 per 100,000 inhabitants, and in the following year it was 59.28. In 2020, there was a significant decline to 34.65, followed by a recovery that continued until peaking in 2023 at 62.47 per 100,000. In 2024, there was another reduction to 55.25.

The study of mortality trends in women showed a slight, non-significant reduction, with an APC of -1.4 (95% CI: -15.1; 14.6) and lower relative variability. The rates were similar to those in the total and male analyses, with figures of 16.43 and 16.82 in 2018 and 2019, respectively; they fell to a low of 7.14 in 2020; recovered to 16.33 in 2023, and fell to 11.40 the following year.

During the period 2018–2024, trend analysis using JoinPoint revealed heterogeneous patterns in mortality from unintentional injuries by age group and gender in Ecuador (Table 1).

Table 1. Joinpoint analysis of mortality from unintentional injuries by age group and gender.

MALES					
			Trend		
Age groups	**2018 Rates**	**2024 Rates**	**Period**	**APC (95% CI)**	***p*-value**
0-14	13.2	9.12	2018-24	-6.7* (-13.5; -0.1)	0.004
15-24	55.1	65.3	2018-24	3.4 (-2.4; 9.6)	0.195
25-44	65.2	72.7	2018-24	4.2 (-1.6; 11.1)	0.162
45-64	76.5	65.4	2018-24	-0.3 (-8.6; 9.3)	0.949
≥65	165.2	94.7	2018-24	-4.5 (-16.3; 9.1)	0.482
FEMALES					
			Trend		
Age groups	**2018 Rates**	**2024 Rates**	**Period**	**APC (95% CI)**	***p*-value**
0-14	9.2	6.2	2018-24	-5.2 (-5.2; 7.5)	0.383
15-24	9.3	10.1	2018-24	2.7 (-5.7; 11.9)	0.522
25-44	9.7	12.6	2018-24	5.3 (-1.1; 12.2)	0.091
45-64	13.8	10.3	2018-24	-1.1 (-13.4; 13.2)	0.861
≥65	83.5	28.6	2018-24	-5.8 (-32.5; 31.6)	0.735

*The Annual Percent Change (APC) is significantly different from zero at the alpha = 0.05 level.

In men, a significant decrease was observed only in the 0–14 age group, with an APC of -6.7 (95% CI: -13.5; -0.1), while no statistically significant changes were detected in the other age groups. In the 15–24 age group, there was an upward but not significant trend with an APC of 3.4 (95% CI: -2.4; 9.6), and similarly in the 25–44 age group

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with an APC of 4.2 (95% CI: -1.6; 11.1), with rates increasing from 55.1 to 65.3 and from 65.2 to 72.7 per 100,000 inhabitants, respectively, between 2018 and 2024. In contrast, the 45–64 and ≥65 age groups experienced non-significant reductions (APC: −0.3 and −4.5, respectively).

In women, none of the age-group trends were statistically significant. However, a decrease in the trend was observed in the 0-14 age group with an APC of -5.2 (95% CI: -5.2; 7.5) and in the over-65 age group with an APC of -5.8 (95% CI: -32.5; 31.61). In contrast, the 15-24 age group showed a slight increase with an APC of 2.7 (95% CI: -5.7; 11.9), as did the 25-44 age group with an APC of 5.3 (95% CI: -1.1; 12.2).

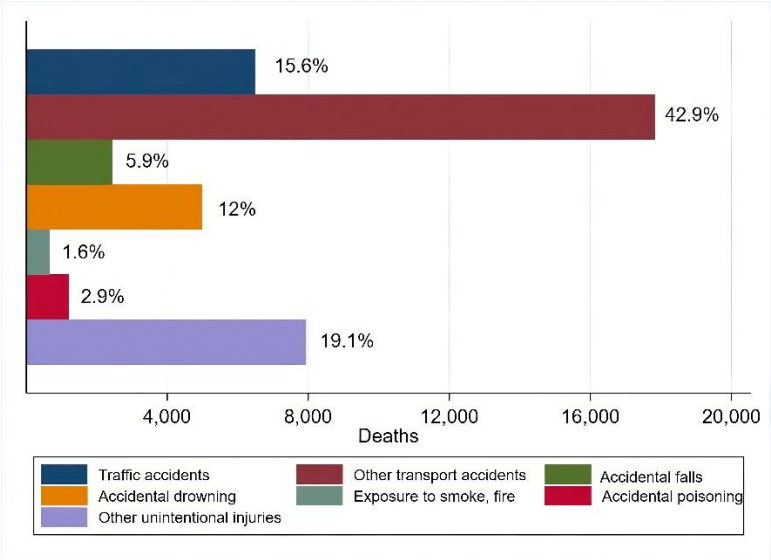


Figure 2. Percentage distribution of mortality by type of unintentional injury

Figure 2 shows the distribution of the main causes of death from unintentional injuries in Ecuador. Transport-related events accounted for the largest share, representing 58.5% of total deaths. Within this group, other transport accidents were the leading cause, accounting for 42.9% (17,835 deaths), followed by other types of accidents at 19.1% (7,935 deaths) and traffic accidents at 15.6% (6,494 deaths). In fourth place were drowning, asphyxiation, and mechanical suffocation, accounting for 12.0% (4,989 deaths). Accidental falls accounted for 5.9% (2,452 deaths), while the least frequent causes were accidental poisoning with 2.9% (1,214 deaths) and accidents involving fire, smoke, and hot substances with 1.6% (672 deaths).

Table 2. Mortality trends for the leading causes of unintentional injuries

MALES					
			Trend		
Unintentional injury	**2018 Rates**	**2024 Rates**	**Period**	**APC (95% CI)**	***p*-value**
Transportation accidents	12.3	6.7	2018-24	-9.5* (-16; -2.5)	<0.001
Falls	5.5	0.1	2018-24	-27.1 (-65.8; 55.5)	0.451
Drowning	7.8	3.9	2018-24	-9.2* (-13.8; -4.2)	<0.001
Poisoning	2.4	0.8	2018-24	-6.6 (-41; 47.5)	0.746
FEMALES					
			Trend		
Unintentional injury	**2018 Rates**	**2024 Rates**	**Period**	**APC (95% CI)**	***p*-value**
Transportation accidents	2.5	1.3	2018-24	-10.7* (-17.9; -2.5)	0.009
Falls	1.5	0.03	2018-24	-26.1 (-68.8; 77.3)	0.534
Drowning	2	0.8	2018-24	-11.7* (-21,1; -0.8)	0.035*
Poisoning	0.4	0.2	2018-24	-3.7 (-29.1; 31.7)	0.808

Mortality from transport accidents declined significantly in both genders during 2018–2024. In men, adjusted rates fell from 12.3 per 100,000 population in 2018 to 6.7 in 2024, with an APC of -9.5 (95% CI: -16; -2.5), indicating a statistically significant reduction. Similarly, rates among women decreased from 2.5 to 1.3 per 100,000, with an APC of -10.7 (95% CI: -17.9 to -2.5), which was statistically significant (Table 2).

Drowning, asphyxiation, and mechanical suffocation showed a significant decline in both sexes. In men, the rate fell from 7.8 in 2018 to 3.9 in 2024 with an APC of -9.2 (95% CI: -13.8; -4.2), while in women it fell from 2.0

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to 0.8 with an APC of -11.7 (95% CI: -21.1; -0.8) (Table 2).

About accidental falls, no statistically significant changes were detected in either gender. In men, the rate decreased from 5.53 in 2018 to 0.1 in 2024 (APC -27.1%; 95% CI: -65.8; 55.5), and in women from 1.5 to 0.03 (APC -26.1%; 95% CI: -68.8; 77.3), with very wide confidence intervals that include both significant increases and decreases, reflecting high variability and the absence of a clear trend. Accidental poisonings also showed no significant trends. In men, the rate fell from 2.4 to 0.8 (APC: -6.6; 95% CI: -41; 47.5), and in women from 0.4 to 0.2 (APC: -3.7; 95% CI: -29.1; 31.7) (Table 2).

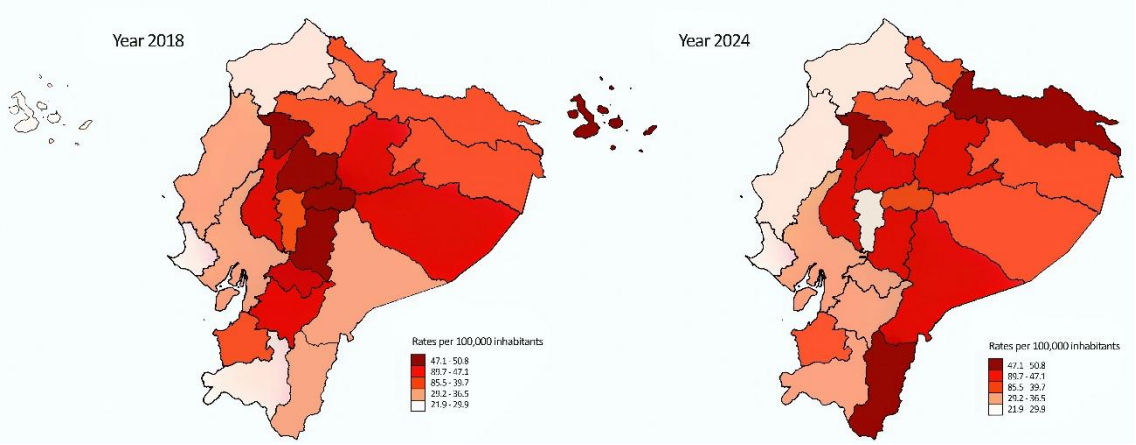


Figure 3. Maps of mortality from unintentional injuries by province in 2018 and 2024.

In 2018, deaths from unintentional injuries were mainly concentrated in provinces in the center of the country, such as Tungurahua with a rate of 50.86 per 100,000 inhabitants, Chimborazo (50.59), Santo Domingo de los Tsachilas (49.59), and Cotopaxi (48.02), followed by Cañar (47.01) and Los Ríos (43.70). In contrast, some provinces in the coastal and Amazon regions showed intermediate or low levels, such as Guayas (29.37) and Esmeraldas (28.03). Galápagos, meanwhile, had the lowest rate in the country, with 21.99 per 100,000 inhabitants (Figure 3).

In 2024, the geographic distribution of unintentional injury mortality was reconfigured, with the areas of highest risk shifting to the Insular and Amazon regions. Galápagos topped the national statistics with a rate of 76.73 per 100,000 inhabitants, followed by Zamora Chinchipe (63.16) and Sucumbíos (54.58). While provinces such as Santo Domingo de los Tsachilas (48.27) and Chimborazo (47.59) maintained high levels, Manabí was at the opposite end of the spectrum, recording the lowest mortality rate in the country (22.00) (Figure 3).

4.DISCUSSION

The trend in mortality from unintentional injuries showed a marked decline in 2020, coinciding with the onset of the COVID-19 pandemic and the implementation of lockdown measures. This finding is consistent with previous research; for example, Holguín-Carvajal et al.[10] reported a 31% reduction in traffic accident deaths compared to 2019. A similar pattern was documented in Peru, characterized by a sharp drop in traffic accident mortality during lockdown, followed by a sustained increase in subsequent years [15]. The recovery observed until 2023, and the decline recorded in 2024, are in line with the "rebound" patterns described in other international contexts following the lifting of mobility restrictions [16,17].

The gender analysis showed that mortality from unintentional injuries affects men to a greater extent, accounting for approximately 80% of cumulative deaths, with a male-to-female ratio of 3.8:1. This pattern is consistent with recent studies, which indicate that, globally, men have mortality rates two to three times higher than women[18,19]. Høye and Hesjevoll [20] mention that one possible explanation for this trend is the greater participation of men in dangerous jobs, driving motor vehicles, risky recreational activities, and behaviors associated with recklessness or alcohol consumption.

The mortality trend by age group showed a reduction in both sexes in the 0-14 age group; however, this decline was statistically significant only in males. This coincides with the steady declines in child injury mortality reported in the United States [21], which may reflect global advances in child safety. Increases in mortality rates from unintentional injuries were observed in young women and men aged 15 to 44. Although these increases were not statistically significant, they are consistent with international evidence indicating a higher risk associated with exposure behaviors and behavioral patterns characteristic of this age group [21,22]. These findings underscore the need for close epidemiological surveillance and the implementation of prevention strategies targeting young males[23].

In contrast, in adults aged 65 years and older, although the present series shows a decrease in mortality, the burden remains high. This result contrasts with the findings of Zhang et al.[24] and Ji et al.[25], who report that deaths from falls in the elderly have shown an upward trend, emphasizing the importance of strengthening safety and prevention policies in this vulnerable group.

Within the global pattern in which transport events are the leading cause of unintentional injury mortality in Ecuador, this study's findings are consistent with international evidence. Research such as that conducted

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by Kettunen et al.[26] and Zaki et al.[27] supports this observation by noting that traffic-related and other transportation accidents are the main causes of mortality across contexts. This predominance is largely explained by structural and social factors characteristic of low- and middle-income countries, such as deficiencies in road infrastructure, limited regulation and enforcement of traffic rules, and greater exposure of the young and economically active population to mobility-related risks [2,28,29].

The causes of mortality from falls, drownings, and poisonings ranked second in terms of the burden attributable to unintentional injuries. These results are consistent with previous studies, including those by Xing et al.[30], which determined that the leading cause is traffic accidents and other transport-related events, while drownings, burns, and falls rank second. In children, the pattern continues: transport injuries are the most common, followed by drownings and then burns [22].

The geographic distribution of mortality from unintentional injuries in Ecuador showed a significant shift between 2018 and 2024, shifting from an initial concentration in the central and southern highlands to a higher burden in the Amazon and island provinces. This is consistent with evidence indicating that regions with lower income, lower literacy rates, and poorer road infrastructure and working conditions have higher mortality rates from injuries, particularly traffic and occupational injuries [10,32]. This shift could be associated with recent road expansion, increased motorcycle use, and greater exposure to high-risk extractive and occupational activities, phenomena observed in other middle-income countries [10,33].

In this type of analysis, it is important to recognize that the figures for deaths from unintentional injuries may be underestimated. This may occur due to errors in coding causes of death under the ICD-10 classification, which may affect the accuracy of the results. Hua et al.[34] and Paleczny et al.[35] point out that estimates of mortality from traffic accidents, falls, drownings, and poisonings may be affected by the use of nonspecific codes. Therefore, these results should be interpreted with caution, particularly when making temporal comparisons with other countries, as discrepancies in coding quality can lead to significant biases in trend estimates.

5.CONCLUSION

The trend in mortality from unintentional injuries in Ecuador between 2018 and 2024 showed a non-significant increase, with a decrease in 2020 linked to COVID-19 pandemic-related restrictions. Mortality affects men more, with a male-to-female ratio of 3.8. In the age-group analysis, a statistically significant decrease was observed among children. Transportation accidents continue to be the leading cause of death, although a notable decrease is being observed. The analysis by province revealed that mortality increases from the mountainous regions to the Amazon and island provinces, indicating territorial inequalities in working conditions, socioeconomic conditions, and road conditions.

Limitations

The main limitation of the study is the possible underestimation of deaths from unintentional injuries, which may be due to errors in the initial coding of causes of death according to ICD-10. In addition, the use of nonspecific or “catch-all” codes for certain types of falls, traffic accidents, or poisonings may affect the accuracy of the estimated rates. Although a provincial analysis was conducted, additional research at the cantonal level is needed to identify smaller geographic clusters with high mortality rates. Future research should focus on prospective multicenter studies that link mortality registries with clinical emergency department data.

Ethical Approval

This study used anonymized secondary data from Ecuador's National Institute of Statistics and Censuses (INEC). Ethical approval was not required, as no human subjects were directly involved.

Consent for publication

Not applicable. This study does not contain any person's data in any form.

Conflict of Interest

The authors declare no conflict of interest.

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This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Availability of data and materials

The data analyzed in this study are publicly available on the official website of Ecuador’s National Institute of Statistics and Censuses (INEC). These statistical records are open access and can be downloaded directly from the INEC’s official website at: https://www.ecuadorencifras.gob.ec/defunciones-generales/

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